

Exercise 5: Task Management System

Scenario

You are developing a task management system where tasks need to be added, deleted, and traversed efficiently.

1. Understand Linked Lists

What is a Linked List?

A Linked List is a linear data structure in which elements, called nodes, are connected using pointers (references). Unlike arrays, linked lists do not store elements in contiguous memory locations, allowing dynamic memory allocation.

Types of Linked Lists

Singly Linked List

- Each node contains data and a reference to the next node.
- Traversal is possible only in the forward direction.
- Uses less memory than a doubly linked list.

Doubly Linked List

- Each node contains data, a reference to the previous node, and a reference to the next node.
- Supports traversal in both forward and backward directions.
- Requires additional memory for storing the previous reference.

2. Setup

A Java project named TaskManagementSystem was created.

The project contains a code folder with the following files:

TaskManagementSystem

└── code

 └── Task.java

 └── TaskManagementSystem.java

- Task.java – Defines the Task class with attributes taskId, taskName, status, and a reference to the next node.

- **TaskManagementSystem.java** – Implements a Singly Linked List and provides operations to add, search, traverse, and delete tasks.

3. Implementation

Task Class

The Task class contains the following attributes:

- **taskId**
- **taskName**
- **status**
- **next**

Linked List Operations

A Singly Linked List is used to manage the tasks.

The following operations are implemented:

- **Add Task**
- **Search Task**
- **Traverse Tasks**
- **Delete Task**

4. Analysis

Time Complexity

Operation	Time Complexity
Add Task (at end)	O(n)
Search Task	O(n)
Traverse Tasks	O(n)
Delete Task	O(n)

Advantages of Linked Lists over Arrays

- **Dynamic size; memory is allocated as needed.**
- **Efficient insertion and deletion without shifting elements.**

- Better memory utilization when the number of elements changes frequently.
- Suitable for applications where frequent insertions and deletions are performed.

5. Conclusion

The Task Management System was successfully implemented using a Singly Linked List. The application supports adding, searching, traversing, and deleting tasks efficiently. Compared to arrays, linked lists provide greater flexibility for dynamic data, making them well-suited for task management applications where records change frequently.